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Research Article

Surface Damage Detection and Localization for Bridge Visual Inspection Based on Deep Learning and 3D Reconstruction

Youhao Ni,[1] Jianxiao Mao,[1] Hao Wang,[1] Zhuo Xi,[1] and Zhengyi Chen[2]

[1]*Key Laboratory of Concrete and Prestressed Concrete Structures of Ministry of Education, Southeast University, Nanjing 210096, China*
[2]*Department of Civil and Environmental Engineering, The Hong Kong University of Science and Technology, Hong Kong, China*

Correspondence should be addressed to Jianxiao Mao; jianxiao@seu.edu.cn and Hao Wang; wanghao1980@seu.edu.cn

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In the process of infrastructure construction in recent decades, there exist millions of bridges in service that need safety inspection for performance assessment. Currently, computer vision and deep learning-based surface damage detection methods can achieve classification and localization of damages at the image level, but achieving precise localization in three-dimensional space is more challenging. To overcome aforementioned limitations, this study proposes a three-stage method of bridge surface damage detection and localization based on three-dimensional (3D) reconstruction. In stage 1, the UAV flight path planning of the bridge is carried out, and the 3D reconstruction model of the bridge is formed based on the structure from motion (SfM) algorithm. In stage 2, you-only-look-once version 7 (YOLOv7) network is adopted to detect multiple damages, and scale invariant feature transform (SIFT) detector is used to match the identical damage in image level. In stage 3, based on solution of epipolar geometric constraint, the matched damage was mapped to the 3D model, and the 3D damage localization is realized and visualized. The quality of the 3D model has been analyzed, and it is recommended that inspection distance is determined at 20 m. Moreover, the reconstruction model of bridges achieves centimeter-level positioning accuracy, and the positioning accuracy of damage reaches the meter level. The mapped model effectively showcases surface damages, providing bridge owners with intuitive insights.

1. Introduction

The structural response of a bridge is affected by various factors, including external loads and environmental conditions, rendering them susceptible to various types of damages [1–3]. Surface damages, among these types, constitute a significant proportion, directly and indirectly impacting structural safety and service life [4]. Common types of surface damage on bridges include cracks, corrosion, spalling, pits, and exposed reinforcement [5]. For bridge owners, periodic visual inspections are routine for bridge maintenance purposes. Currently, surface damage detection leveraging computer vision and deep learning has undergone significant advancements, enabling the identification of damages at the image level with high precision [6, 7]. However, due to the wide distribution of damages along bridges, existing methods face challenges in simultaneously detecting and locating surface damages entirely. Moreover, localizing surface damages in the three-dimensional models proves beneficial for effective bridge management, as it provides pertinent information for the accurate decision-making of bridge owners [8, 9]. Therefore, achieving high-precision detection and localization of bridge surface damages in three-dimensional models holds considerable significance [10].

The development of damage detection and localization methods has undergone a certain process, with early emphasis on image-based detection in terms of vision [11]. Image-based damage detection methods generally include damage classification, detection, and segmentation [12]. Damage classification methods typically employ deep convolutional neural networks, which slide convolutional kernels over images to extract local features, thereby achieving category division of image blocks. However, classification methods mainly remain at the image level and can be the

basis for damage detection and segmentation networks by adjusting the output network and adding loss functions for position regression [13]. In contrast to damage classification methods, damage detection methods employ bounding boxes to autonomously locate the damaged parts of bridges while determining the types of bridge damage contained in the images [14]. Damage detection methods include single-stage and two-stage methods, with single-stage methods requiring only one feature extraction to achieve end-to-end damage object detection, thus improving inference efficiency [15–17]. Furthermore, for convenient bridge damage detection, combining unmanned aerial vehicles (UAVs) with single-stage algorithms can replace personnel to conduct rapid inspections of high-risk bridge components such as piers and abutments [18]. To intuitively represent the real size of the damage, during UAV inspections, internal parameters such as camera focal length and pixel offset are generally fixed. Oblique photography can calculate the spatial coordinates of damage considering factors such as camera intrinsic and extrinsic parameters for the same target in continuous frame images obtained from damage detection algorithms, thus providing the possibility for further determination of the damage location [19]. Compared to damage detection methods, although damage segmentation methods can perform per-pixel damage classification, they struggle to establish the uniqueness matching of damage in continuous frame images, thereby limiting their combination with oblique photography for spatial localization of damage.

Generally, image-based damage detection reflects the two-dimensional information of bridge damage. To obtain the three-dimensional spatial information of bridge damage, point cloud data can be acquired to form a three-dimensional model of the bridge over its operational cycle [20]. Based on the results of three-dimensional modeling at different stages, geometric quality inspection of point clouds can be achieved [21]. In addition, some classical point cloud-processing networks can segment objects from point cloud data, such as PointNet [22], DGCNN [23], and Point Transformer [24]. However, despite the routine inspection of point cloud bridges for damages, whether by laser scanning or oblique photography, they are mostly used for global modeling of bridges [25]. Due to the small size of most bridge damage features, it is difficult to extract local damages from point cloud models as inputs for deep learning networks [26]. Currently, research on point cloud-based target detection mostly focuses on the automatic extraction of individual structural components of bridges. Moreover, point cloud data samples are difficult to obtain, and annotated datasets are also small in scale. Furthermore, when measurement devices are far from bridges, the modeling accuracy of point clouds is affected, further reducing the accuracy of point cloud-based target detection. Therefore, adopting image-based surface damage detection methods is more suitable for inspecting local damages. In addition, combining the results of image-based damage detection with 3D point clouds can provide a more feasible approach to locating damages in a three-dimensional space.

To address the research gap, this study proposes a novel approach of surface damage detection and spatial localization based on YOLO network and three-dimensional mapping. Initially, the UAV flight path is meticulously charted based on the generated preliminary model and the 3D reconstruction of the bridge model is realized by the SfM algorithm. Subsequently, leveraging the YOLOv7 object detection algorithm, a comprehensive bridge multidamage detection model is established to identify surface damages within the acquired images. Employing the scale-invariant feature transform (SIFT) detector and the precise pixel-wise coordinates of the damages, seamless image-level matching across the image sequence is achieved. Afterwards, to correlate the damages with the 3D model, solutions involving polar geometric constraints are implemented, followed by a meticulous analysis of positioning accuracy. Ultimately, surface damage information, comprising damage types and corresponding detected images, is seamlessly mapped to the 3D model, providing owners with an intuitive display of bridge damages.

Our primary contributions are as follows:

(1) Propose a spatial localization method for surface damages of bridges, achieving detailed three-dimensional model reconstruction of bridges, with centimeter-level positioning accuracy of the model and meter-level positioning accuracy for damage localization.

(2) Analyze the impact of IoU threshold on the spatial positioning accuracy of damages and propose reasonable IoU threshold recommendations.

(3) Optimize the UAV inspection path by employing the ant colony algorithm and provide suggested parameters comprising flight path, shooting distance, and overlap ratio.

This study is organized as follows. Section 2 reviews related studies on image-based damage detection methods and vision-based damage localization methods. Section 3 describes the proposed technical workflow for surface damage detection and localization, which is implemented in three stages, including 3D reconstruction, image-based damage detection, and damage correlation. Section 4 implements 3D reconstruction of bridges based on UAV and optimizes inspection routes. Section 5 conducts precision analysis of damage detection and localization and visualizes the correlation between damage and the reconstruction model. In closing, Section 5 concludes our study and discusses future research directions.

2. Literature Review

2.1. Image-Based Damage Detection Methods. Currently, surface damage detection methods based on deep learning can be divided into image classification, object detection, and semantic segmentation [27–29]. Among them, object detection algorithms, including single-stage and two-stage methods, are the most widely utilized. For instance, Kim et al. [30] employed region-based convolutional neural

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networks (R-CNNs) using unmanned aerial vehicles (UAVs) to identify cracks on structures, accurately quantifying their thickness and length. Moreover, Faster R-CNN, an evolved version of the two-stage method R-CNN, was employed with database images to detect cracks and spalling on the Anzac bridge surface after an earthquake, achieving an accuracy of 84.7% [31]. To enhance the recognition capability of the network structure, Song et al. [32] proposed a preliminary selection and refinement recognition scheme based on Faster R-CNN, aiming to improve the accuracy of identifying three types of surface damage on bridges, including honeycomb pitting, cracks, and salt erosion. Yu et al. [33] utilized the k-means clustering algorithm to obtain cluster anchor sizes corresponding to the damage in the data, thereby boosting the recognition accuracy of Faster R-CNN with predefined anchors to 84.56%. However, the two-stage algorithm involves the region proposal network (RPN) stage. Although there is an improvement in accuracy, each candidate region undergoes processing within the network, leading to redundant extraction of the same features and thus wasting computational resources.

As single-stage detection algorithms, Single Shot MultiBox Detector (SSD) and You Only Look Once (YOLO) series algorithms have attracted considerable attention. SSD and YOLO discard the RPN stage, thereby achieving fast detection speed while maintaining relatively low accuracy degradation. To further enhance prediction accuracy, Qi et al. [34] introduced feature resolution analysis in SSD. However, due to the relatively shallow network architecture, the degree of improvement is not significant. In contrast, the YOLO series adopt an end-to-end network structure, simultaneously integrating bounding box detection and class confidence score prediction, maintaining a balance between accuracy and inference speed. To achieve detection of bridge cracks, a YOLOv4-FPM deep learning model was proposed based on YOLOv4, optimizing the loss function with focal loss to enhance accuracy and overcome challenges posed by complex backgrounds [18]. Based on YOLO models and transfer learning, a faster and simpler single-stage detector was proposed, capable of recognizing four types of concrete bridge damage, i.e., cracks, spalling, pop-out, and exposed reinforcement [35]. YOLOv7, as a single-stage algorithm, demonstrates superior performance in both recognition accuracy and speed compared to most recent object detectors. Particularly, YOLOv7-E6 outperforms both transformer-based and convolution-based detectors, showing promise for the multidamage detection [36, 37].

2.2. Vision-Based Damage Localization Methods. For a large structural health monitoring, damage localization plays a pivotal role in three-dimensional bridge damage assessment [38, 24]. Based on the point cloud model, Ye et al. [39] proposed a geometric deformation-based method for quantifying the damages, accurately pinpointing the shape and direction of damages in masonry arch bridges. Liu et al. [40] utilized a digital surface model to construct a deformation map, identifying potential damage areas of the bridge. By analyzing displacement time series data, bridge damages were assessed using the extreme point symmetric mode decomposition method. Owing to global positioning systems (GPSs) and structure from motion (SfM) algorithm, Chen et al. [21] proposed a bridge damage assessment method based on 3D reconstruction model, and the localization error of spalling is within 25%. Overall, damage localization in point clouds typically employs feature-based or deformation-based processing algorithms, leading to notable differences in accuracy and robustness depending on the chosen algorithm.

Moreover, machine learning methods were introduced for locating damages. Kim et al. [30] introduced a dynamic graph convolutional neural network to identify the damage location in three-dimensional space and classify bridge components. The Mean Intersection of Union (MIoU) value was 86.85%, better than PointNet and PointCNN. Nasrollahi et al. [41] proposed an improved PointNet-based method for bridge surface damage recognition, collecting a dataset with 3572 labeled samples for damage and nondamage classification. Based on the multiview classification method, Isailovi et al. [42] detected the spalling damage from the point clusters, which were divided from the point cloud model. However, the point cloud datasets for bridge damages are currently relatively limited, which constrains the improvement of detection model accuracy. Meanwhile, certain damages such as cracks occupy a small proportion within the point cloud model, and their features are not easily extracted, further impacting the precision of damage localization [26, 43].

Compared with the above point-cloud-model-based methods, the image-based damage detection methods possess certain advantages, including high precision, available datasets, and real-time capabilities [44, 45]. For small damages, the high resolution of images enhances the robustness of damage feature extraction. Hence, detecting surface damage at the image level and subsequently associating the pixelwise coordinates of damages with the 3D model is a rational approach [29]. Perry et al. [46] developed a damage mapping technology that associate damages in 2D images with 3D point cloud models, employing machine-learning-based methods for detection. Liu et al. [9] presents an image-based bridge pier crack assessment method using deep learning and 3D scene reconstruction, projecting the cracks onto a 3D surface triangular model. Such methods projected images into the point cloud triangulated grid for damage location, and the accuracy depends on model updating. Furthermore, based on two-dimensional Delaunay triangulation, Won et al. [47] proposed a depth-matched panoramic image stitching method to reconstruct the continuous crack path. Yoon et al. [48] proposed a method of interest region detection based on three-dimensional image coordinates, which utilizes monocular camera and one-dimensional LiDAR for data fusion. Jiang et al. [7] developed a lightweight visual guidance UAV system utilizing CNN to detect and locate cracks, spallation, and corrosion, integrating vision-inertial fusion sensors and ultrasonic rangefinders for positioning. However, in the localization process, existing

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studies require a continuous image series or the introduction of auxiliary parameters for positioning, such as radar ranging.

Deep learning-based methods can accurately locate the pixelwise coordinates in images, while the 3D location remains to be further studied. For bridge owners, combining images and 3D models can enhance detection accuracy and intuitively demonstrate damage spatial location, aiding maintenance. To achieve three-dimensional spatial localization of damage, this paper proposes a three-stage approach for three-dimensional damage localization, leveraging the high precision of image-based damage detection and 3D spatial localization advantages.

3. Methods of 3D Reconstruction and Surface Damage Localization

3.1. Workflow of the Proposed Method. The workflow of surface damage localization consists of three phases, i.e., 3D reconstruction, deep learning-based damage detection, and three-dimensional damage mapping, as shown in Figure 1. Phase I is the process of generating a 3D reconstruction model of bridges based on Structure from Motion (SfM) [49]. RTK coordinates, comprising longitude, latitude, and altitude, serve as the input for damage mapping. In phase II, surface damages of bridges are detected using a deep learning model, and identical damages from successive images are associated through feature matching algorithms. Phase is the process of three-dimensional damage mapping. Based on the solution of epipolar geometric constraint [50], the 3D model of bridges is formed with damage localization information.

3.2. Phase: 3D Reconstruction of the Target Bridge. In phase, the primary objective of the methodology is to conduct 3D reconstruction of the target bridge. Initially, the UAV platform is carefully chosen, and UAV path planning is meticulously executed. Subsequently, camera posture information, comprising pitching angle, roll angle, and course angle, is captured by RTK module. Moreover, spatial coordinate information of the camera is also acquired, including longitude, latitude, and altitude. Ultimately, leveraging the Structure from Motion (SfM) algorithm, the 3D reconstruction model of bridge is generated.

The detailed workflow of 3D reconstruction consists of three stages [51], as shown in Figure 2. In the UAV path planning stage, preliminary route planning is essential to generate a rough 3D model. Subsequently, detailed flight path planning is conducted based on this rough model. The UAV captures the required photos for fine 3D model reconstruction along the detailed flight path. By capturing images from various viewpoints, Structure from Motion (SfM) utilizes multiple 2D image views to construct the 3D geometry, i.e., the structure, of a scene. Key point features, extracted from multiple 2D images, are interconnected and assembled to form a cohesive spatial 3D structure [52, 53]. In the SfM stage, camera position is estimated by extracting features and matching features. The "keypoints filtering" step is utilized to filter out keypoints that are low in robustness and susceptible to noise interference [54, 55]. Subsequently, the "mismatched points filtering" step is employed to eliminate incorrectly matched keypoints [56–58]. Sparse point clouds are then generated, followed by model parameter calibration to produce dense point clouds.

In the process of 3D reconstruction, a series of model processing operations are performed, including point cloud processing, building mesh, building texture, and building 3D model [59, 60]. For multiview stereo computation and reconstruction in this study, the 3D reconstruction software ContextCapture, developed by Bentley Systems, was selected.

3.3. Phase: Deep Learning-Based Damage Detection. YOLOv7, a single-stage object detection network, is the official version optimized and improved on YOLOv5 [36]. The network consists of three parts, comprising backbone, neck, and head. Within the backbone, an extended efficient layer aggregation network is introduced to facilitate feature extraction, while an auxiliary head is integrated to enhance training efficiency. YOLOv7 effectively manages the tradeoff between parameters, computational complexity, and performance. On the Microsoft Common Objects in Context dataset, YOLOv7 achieves average precision (AP) of 56.8% on GPU V100, marking the highest performance among real-time detection algorithms, with a frame rate surpassing 30 frames per second [36].

YOLOv7 network is employed for damage localization at the image level. According to the research, to generate a reasonable 3D model, it is imperative that the forward overlap and side overlap rates exceed 80% and 70%, respectively [61]. Feature matching entails establishing a one-to-one correspondence between consecutive images obtained through oblique photography. Consequently, identical damage observed in two consecutive images are linked, thereby laying a foundation for calculating the spatial coordinates of the damage. The scale-invariant feature transform (SIFT) detector serves as a local feature detection algorithm [62]. The detector processes feature point matching between images and obtains the scale and orientation descriptor of interest points or corner points [63].

Based on the SIFT detector, damage feature matching is illustrated in Figure 3. Figure 3(a) illustrates the keypoint detection results by the SIFT detector. Following the "Keypoints filtering" step, Figure 3(b) is derived, which removes nonhigh-quality keypoints lacking significance and susceptible to noise interference. In the "Keypoints filtering" step, both the method based on gradient magnitude and orientation for keypoint selection and filtering out small keypoints are employed [64]. The concrete patch of the bridge pier, serving as identical damage across two consecutive images, is detected using the multidamage model of YOLOv7. Figure 3(c) shows the results of keypoint matching using SIFT descriptors, revealing the presence of some mismatched points. For the "mismatched points filtering" step, the method involving the RANSAC algorithm and ratio test is chosen for filtering, resulting in the correct matching

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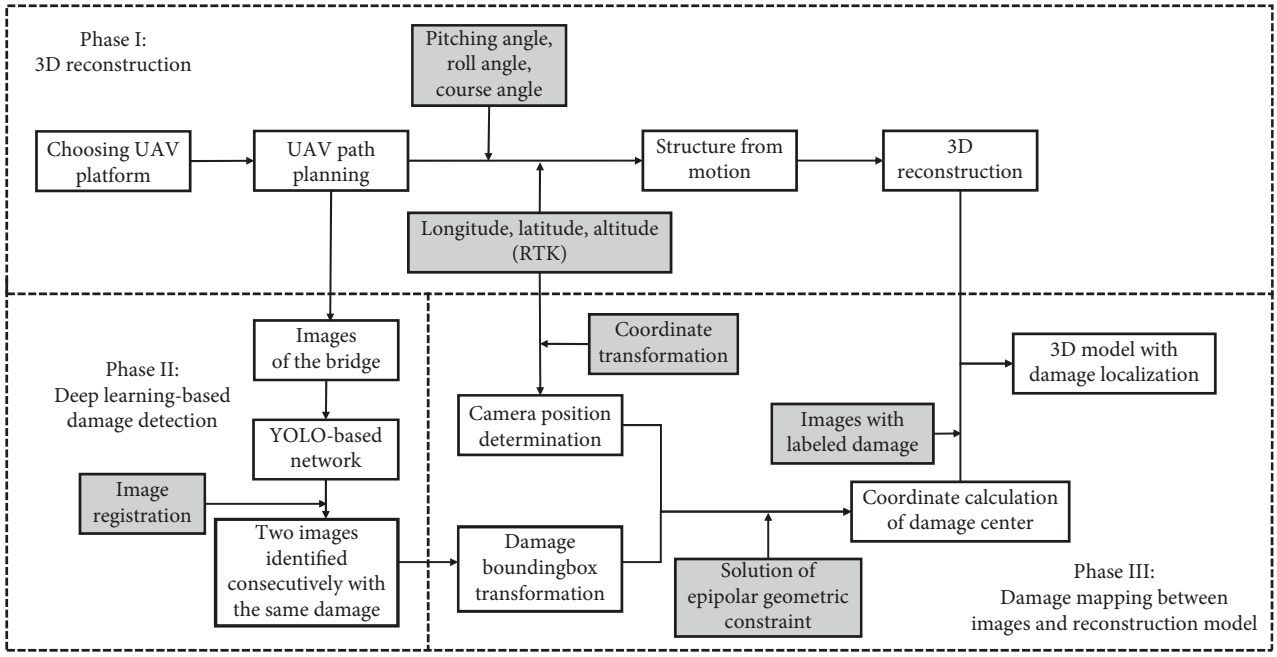


Figure 1: Workflow of surface damage localization.

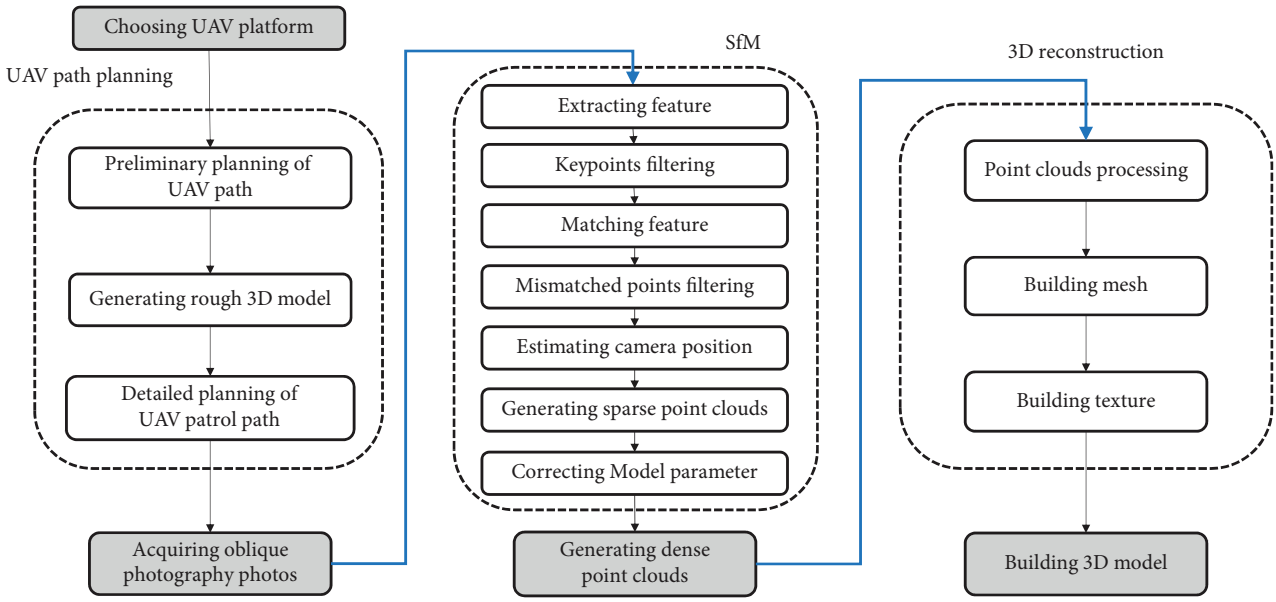


Figure 2: Detailed workflow of 3D reconstruction of the target bridge.

results depicted in Figure 3(d) [65]. Due to varying shooting angles, the bounding boxes of damages vary. Utilizing the SIFT detector, four feature points of the identical damage detection box are correlated.

3.4. Phase: Damage Mapping between Images and the Reconstruction Model. The relationship between the 3D space and the image can be established through by the pinhole imaging principle [66]. The principle is detailed in a coordinate transformation diagram of camera vision system, depicted in Figure 4.

The system contains four coordinate systems, including the world coordinate system, the camera coordinate system, the image physical coordinate system, and the image pixel coordinate system. The transformation relationships between the coordinate systems are delineated as follows:

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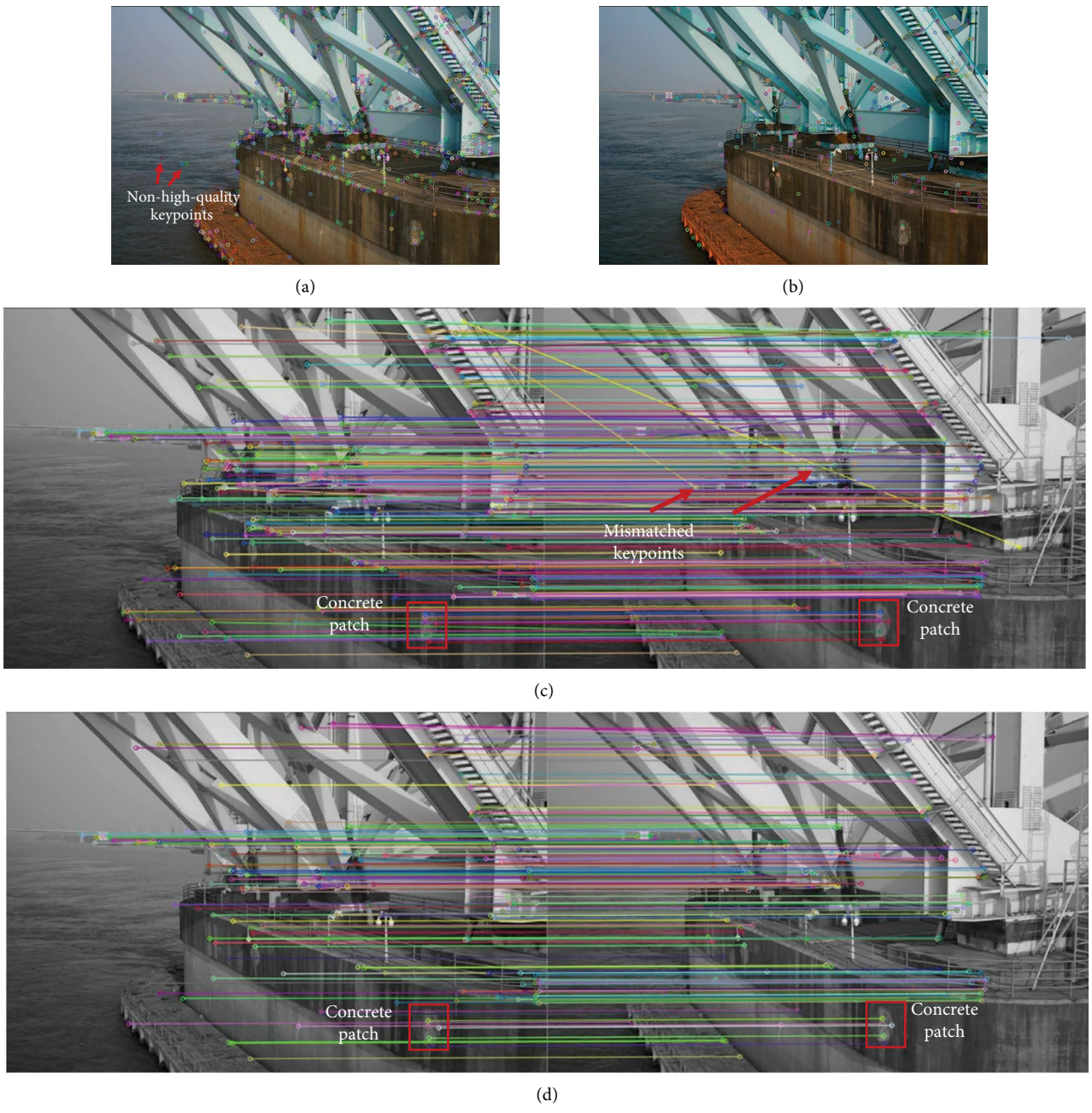


FIGURE 3: Damage feature matching based on SIFT detector. (a) Before keypoints filtering. (b) After keypoints filtering. (c) Before mismatched points filtering. (d) After mismatched points filtering.

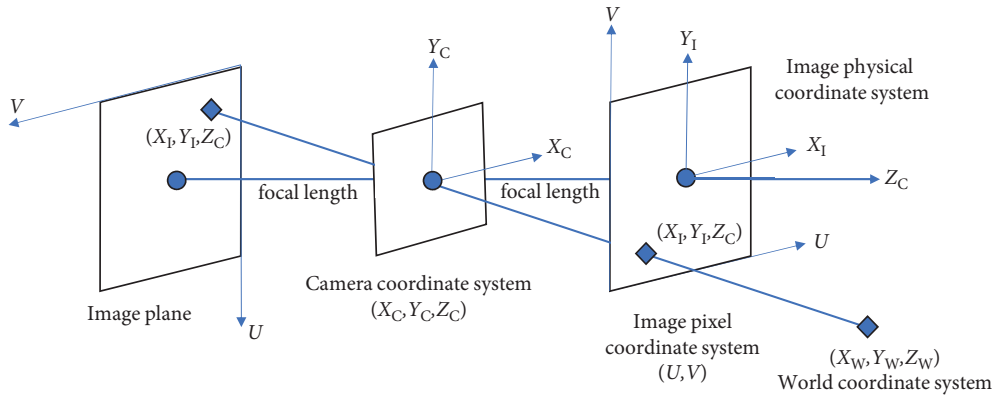


FIGURE 4: Coordinate transformation diagram of camera vision system.

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$$\begin{bmatrix} X_C \\ Y_C \\ Z_C \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R} & \mathbf{T} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} X_W \\ Y_W \\ Z_W \\ 1 \end{bmatrix}$$
$$\mathbf{R} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}, \quad (1)$$
$$\mathbf{T} = \begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix}$$

where $[\, X_C \;\; Y_C \;\; Z_C \;\; 1 \,]^{\mathrm{T}}$ and $[\, X_W \;\; Y_W \;\; Z_W \;\; 1 \,]^{\mathrm{T}}$ represent the homogeneous coordinates of the damage in camera and world coordinate system, respectively; $\mathbf{R}$ is the rotation matrix with 3 degrees of freedom; $\mathbf{T}$ is the translation matrix; and $\begin{bmatrix} \mathbf{R} & \mathbf{T} \\ 0 & 1 \end{bmatrix}_{3\times4}$ is the external parameter matrix of the camera.

By internal parameter matrix, the camera coordinate system is converted to the image pixel coordinate system as follows:

$$x = \frac{X_C \times f}{Z_C}$$
$$y = \frac{Y_C \times f}{Z_C} \quad (2)$$
$$Z_C \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_C \\ Y_C \\ Z_C \end{bmatrix},$$

where f denotes the focal length; $[\, x \;\; y \;\; 1 \,]^{\mathrm{T}}$ represents the normalized image physical coordinates. Through the pixel conversion matrix, the image physical coordinate system can be transformed to the pixel coordinate system.

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\mathrm{d}x} & \gamma & u_0 \\ 0 & \frac{1}{\mathrm{d}y} & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}, \quad (3)$$

where γ is the distortion factor; $\mathrm{d}x$ and $\mathrm{d}y$ denotes the pixel length in decimal units, respectively, in x and y directions; and u_0 and v_0 represent pixel coordinate difference between the center and left corner of the image, respectively, in x and y directions.

$$Z_C \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\mathrm{d}x} & \gamma & u_0 \\ 0 & \frac{1}{\mathrm{d}y} & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} [\mathbf{R} \,|\, \mathbf{T}] \begin{bmatrix} X_W \\ Y_W \\ Z_W \\ 1 \end{bmatrix}, \quad (4)$$

where the transformation relationship on the right side of equation (4), from right to left, is the world coordinate system to camera coordinate system, camera coordinate system to image physical coordinate system, and image physical coordinate system to image pixel coordinate system, respectively.

The camera is mounted on UAV, and the relation diagram between target and pixel should be considered in UAV coordinate system, as shown in Figure 5.

Generally, unmanned aerial vehicles (UAVs) utilize the local North-East-Down (NED) coordinate system. It is necessary to determine the transformation relationship between the real target of the camera perspective and the camera coordinate system. Within the camera perspective, the optical axis intersects the image center. The real target is surrounded by damage boundary box, and the center is represented by (X_{WT}, Y_{WT}, Z_{WT}) in the image pixel coordinate system.

The correlation between the pixel point and space point of identical damage is established through 3D mapping, as depicted in Figure 6. The UAV is equipped with a camera to obtain the identical damage from two camera positions. At camera position 1, the camera optical center 1 is connected with the damage bounding box center in pixel wise, to obtain target optical axis L_1. At camera position 2, the camera optical center 2 is connected with the damage bounding box center in pixel wise to obtain optical axis L_2. The intersection coordinate of optical axis 1 and optical axis 2 is (X_W, Y_W, Z_W), which is the real spatial coordinate of the damage.

Regardless of the noise, L_1 and L_2 intersect at the point P_0, which is expressed as

$$\mathbf{P}_0 = \mathbf{L}_1 \times \mathbf{L}_2. \quad (5)$$

The relation between P_1, P_2, and P_0 can be expressed as

$$\begin{aligned} \mathbf{P}_1 &= \mathbf{K}_2 [\, \mathbf{I} \;\; \mathbf{0} \,]_{3\times4} \mathbf{P}_0 \\ \mathbf{P}_2 &= \mathbf{K}_2 [\, \mathbf{R}_{1,2} \;\; \mathbf{T}_{1,2} \,]_{3\times4} \mathbf{P}_0, \end{aligned} \quad (6)$$

$$\mathrm{AP} = \left(\mathbf{K}_2 [\, \mathbf{R}_{1,2} \;\; \mathbf{T}_{1,2} \,] - \mathbf{K}_1 [\, \mathbf{I} \;\; \mathbf{0} \,]\right)_{4\times4} \mathbf{P}_0 = 0, \quad (7)$$

$$\mathbf{A} = \mathrm{UDV}^{\mathrm{T}}, \quad (8)$$

where $\mathbf{K}_1$ and $\mathbf{K}_2$ are the internal parameter matrix of camera I and camera II, respectively; $\mathbf{R}_{1,2}$ is the rotation matrix from camera I to camera II; $\mathbf{T}_{1,2}$ is the translation matrix from camera I to camera II; and $\mathbf{I}$ is the identity matrix.

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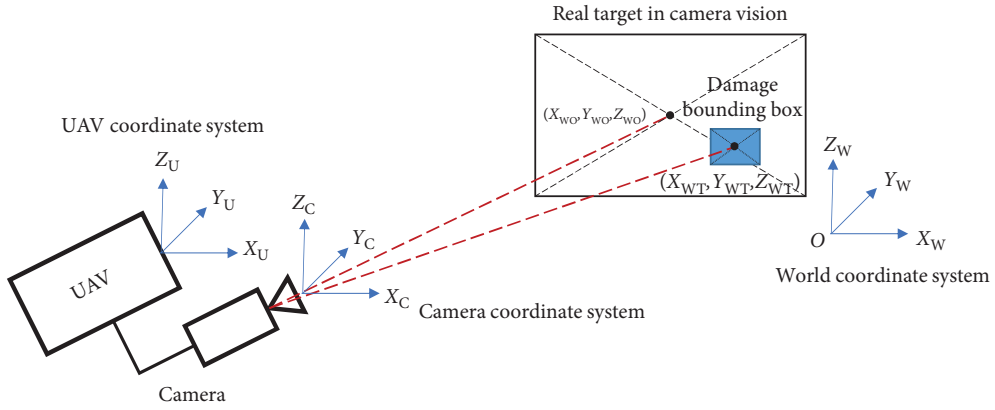


Figure 5: Relation diagram between target and pixel in the UAV coordinate system.

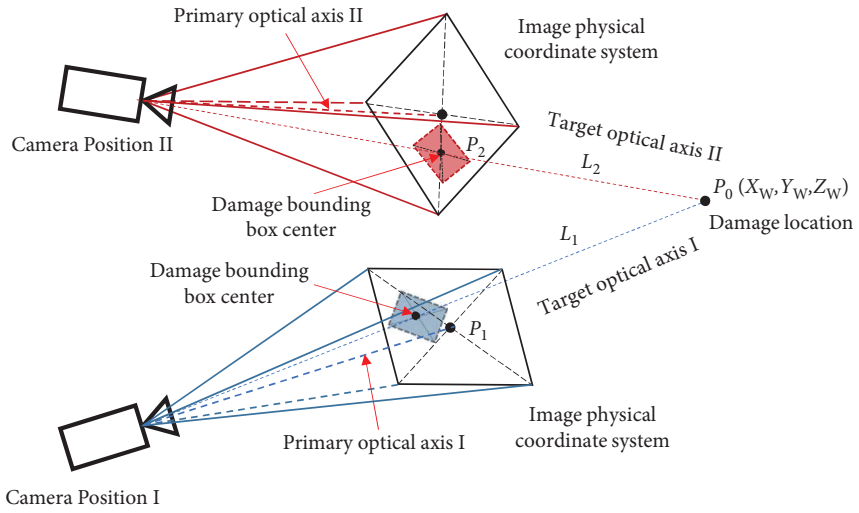


Figure 6: 3D mapping diagram between pixel point and space point.

According to equation (7), the matrix **A** is decomposed by singular value, and the mean values of **U**, **D**, and **V** are obtained, respectively. The last column of the matrix **V** is the spatial homogeneous coordinate of the damage point P_0.

4. 3D Reconstruction Based on UAV Oblique Photography

4.1. Engineering Background. The Dashengguan Yangtze River bridge, a crucial component of the Beijing-Shanghai high-speed railway, is selected as engineering background. Standing as a key infrastructure, the main structure of the bridge is distinguished by its blue print and has been operational for over a decade. In adherence to the maintenance standards of high-speed railway bridges, routine health inspections are currently required, which encompass a surface inspection of the coating. The UAV oblique photography is shown in Figure 7. For this endeavor, the study employs the DJI M300RTK UAV, equipped with ZENMUSE P1 camera boasting a resolution of 6k.

The UAV platform parameters are shown in Table 1. GNSS can use GPS, GLONASS, BeiDou, and Galileo. The position accuracy of RTK is 1 cm + 1 ppm horizontally and 1.5 cm + 1 ppm vertically.

The parameters of the oblique camera are shown in Table 2. The ZENMUSE P1 camera boasts a resolution of 8192 × 5460, a fixed focal length of 35 mm, an aperture of F2.8, and a camera field of view spanning 63.5°.

A total of 10 GCPs (Ground Control Points) and 6 CPs (Checking Points) were set up, located respectively, at the ends of the four piers of the bridge and at three abutment piers. One GCP was set on each side of the abutment piers, and one CP was set on each side of the intermediate piers. The distribution of GCPs and CPs is shown in Figure 8(a).

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(a) (b)

Figure 7: UAV oblique photography. (a) UAV platform. (b) On-site photography.

Table 1: UAV platform parameters.

Item	Parameter
Size	810 × 670 × 430 mm (length × width × height)
Motor wheelbase	895 mm
Maximum tolerable wind speed	12 m/s
Maximum flight altitude	5000 m/7000 m
Maximum pitch angle	30°
Obstacle sensing range	0.7–40 m (horizontally) 0.6–30 m (vertically)
GNSS	GPS + GLONASS + BeiDou + Galileo
RTK position accuracy	1 cm + 1 ppm (horizontally) 1.5 cm + 1 ppm (vertically)

Table 2: Oblique camera parameters.

Item	Parameters
Production name	ZENMUSE P1
Focal length	35 mm/F2.8
Resolution	8192 × 5460
Field of view	63.5°
Type of sensor	CCD
Pixel size	4.4 μm
Working temperature	−20°C–50°C

First, target markers were designed and deployed on-site and then GCPs and CPs were accurately measured using an RTK (real-time kinematic) marker, as shown in Figure 8(b). RTK markers can achieve centimeter-level positioning accuracy.

4.2. UAV Path Planning

4.2.1. Preliminary Flight Path. To generate the preliminary bridge model for fine route planning, the aerial photography path is designed based on two-dimensional plane, as depicted in Figure 9(a). DJI GS PRO was employed for flight path planning. However, due to high-speed railway bridge management regulations, UAVs are restricted from flying over the bridge deck. Consequently, we employ equally spaced parallel lines to create an optimal S-shaped flight path. The shooting distance interval during the flight is determined according to the image overlap rate.

4.2.2. Fine Path Planning of the Whole Bridge. Based on ContextCapture, the preliminary model is generated. The flight path is determined based on the flight direction, height, and shooting distance. For flight path planning, WPM Track Master 3.9 is employed, and the software offers various autonomous path planning modes such as waypoint flight and fine inspection. This study plans the three-dimensional route independently according to the needs of the oblique photography reconstruction.

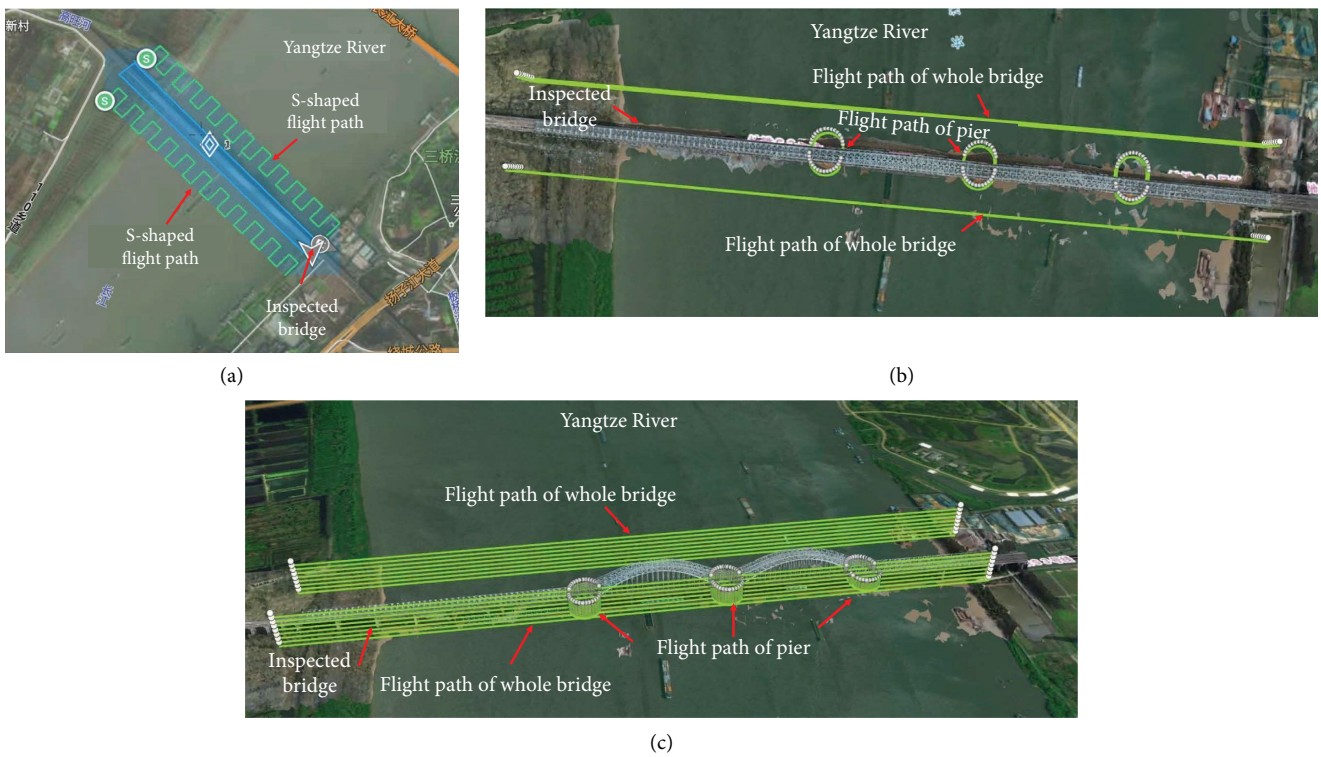
The UAV fine flight path is designed for the whole bridge, as shown in Figure 8. The plan and elevation of the fine flight path are shown in Figures 9(b) and 9(c), respectively. The UAV takes off from the upstream side of the Dashengguan Bridge, conducts in close proximity along both sides of the upstream and downstream, and collects data by spiraling around each pier, and returns after completion. When capturing the entire bridge, there are four routes on each side in elevation. Meanwhile, supplementary flight shooting is conducted at the top of the bridge arch. Considering the operation safety and the reconstruction accuracy, the UAV shooting distance is at 20 m.

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(a) (b)

Figure 8: Distribution of GCPs and CPs. (a) Location marks. (b) Coordinates measurement.



(a) (b)

(c)

Figure 9: The UAV fine flight path of whole bridge. (a) Two-dimensional flight path planning. (b) The flight path plan. (c) The flight path elevation.

4.2.3. Fine Path Planning of the Bridge Pier. The pier dimension is about 11 m × 27 m in plane size, with a height above the water standing at about 30 m. The pier is simplified into cuboid and the cuboid spiral flight path is optimized. This optimization aims to ensure continuous camera angle coverage of the entire pier surface, without factoring in time losses due to UAV hovering and turning.

In Figure 10(a), the plane view of the UAV path is depicted, featuring point A as the starting point and point B as an intermediate point traversed by the UAV, both located at the diagonal points of the bridge pier. The ant colony algorithm was employed, and the population and iteration number were set to 20 and 500, respectively. In addition, the maximum lateral and vertical variation distance of ants and the pheromone decay were set to one unit and 0.9,

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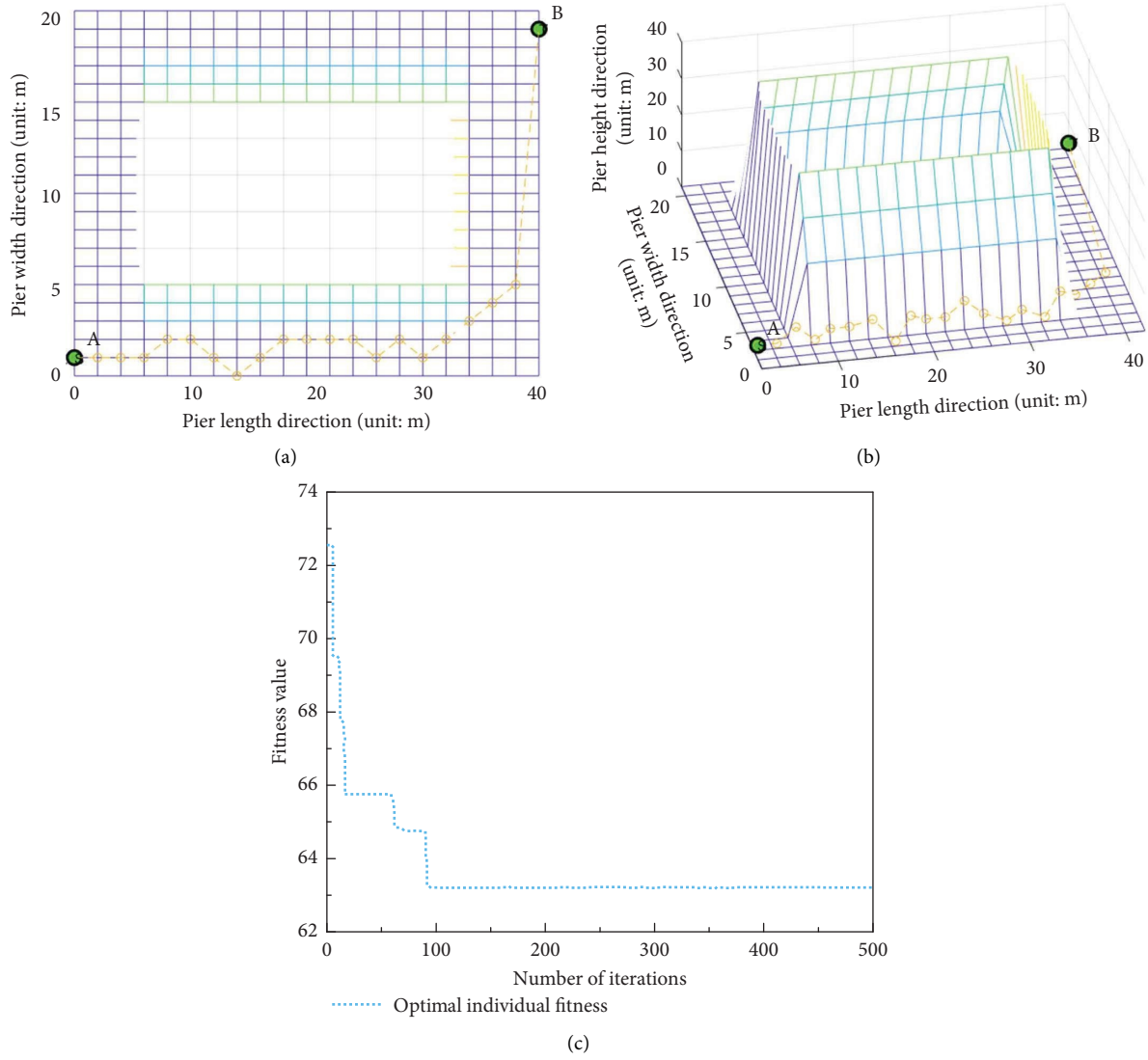


FIGURE 10: Fine path planning of the bridge pier. (a) Plane view of the UAV path. (b) Spatial view of the UAV path. (c) Trends of the optimal individual fitness.

respectively. The trends of the optimal individual fitness are shown in Figure 10(c). As the iteration number approaches 100, the fitness value reaches the minimum and converges. Figures 10(a) and 10(b) are the plane view and spatial view of UAV path, respectively. Therefore, to achieve the shortest flight path, the UAV should fly in a spiral shape around the pier.

Based on the abovementioned analysis, the WPM track Master was employed to plan the flight path. A top-down spiral route was designed to acquire a comprehensive shot of the pier, and the flight speed was within 1 m/s, and the P1 camera utilizes a mechanical shutter with a speed of 1/2000 second. To assess the influence of various flight distances and overlap rates on the accuracy of the 3D model, a total of five flight paths were planned, as detailed in Table 3.

Route 1, route 2-3, and route 3 flew spiral at the distance of 10 m, 20 m, and 30 m from the pier, and the forward and side overlap rates were 80% and 70%, respectively. The shooting distance of route 2-1 and route 2-2 is 20 m, the forward and side overlap rates of route 2-1 is 60% and 50%, and the forward and side overlap rates of route 2-2 is 70% and 60%, respectively. The fine flight path of the bridge pier is shown in Figure 11. The camera pitch angle is −5°, that is, the overhead shot angle is 5°.

4.3. Quality Analysis of the Reconstruction Model. ContextCapture software was used to reconstruct the model. After triangulation, dense point cloud, grid generation, and texture generation, the detailed model of the whole bridge was obtained. During the matching stage, the keypoint density is configured as normal, the optical attribute evaluation mode is set to one step, and parameters such as focal length, principal point, distortion, and aspect ratio are adjusted using the least squares method. In the reconstruction stage, the tile size is set to 80 meters, a regular planar grid is

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Table 3: Pier route planning.

Path ID	Flight distance (m)	Forward/side overlap (%)	Camera pitch angle	Designed ground sampling distance (cm/px)
Route 1	10	80/70	−5°	0.321
Route 2-1	20	60/50	−5°	0.486
Route 2-2	20	70/60	−5°	0.448
Route 2-3	20	80/70	−5°	0.388
Route 3	30	80/70	−5°	0.724

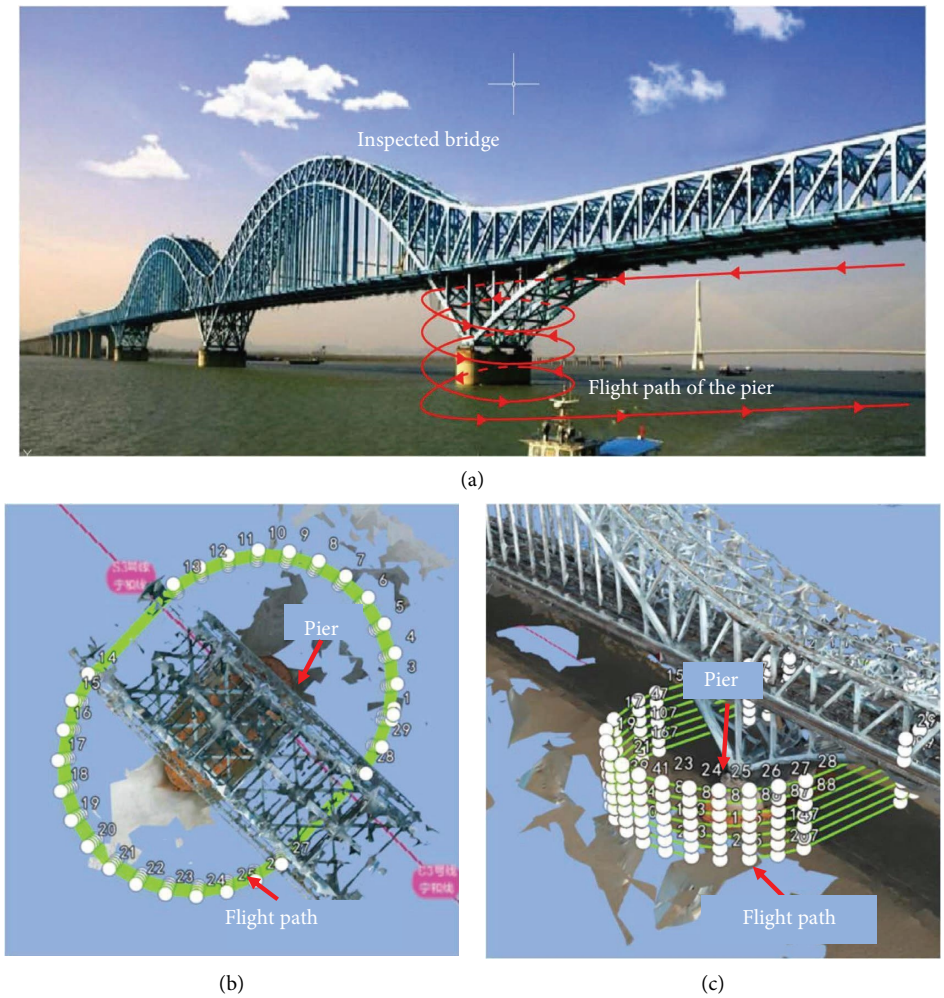


Figure 11: Fine flight path of the bridge pier. (a) Schematic drawing. (b) Plan view of the flight path. (c) Spatial view of the flight path.

selected, and hole filling is performed with the option "fill all holes except tile boundaries." To measure the accuracy of the model location, six checking points (CPs) were placed near bridge piers, with two checking points located at the piers in the river and four on the piers at the shore [67]. For each checking point, precise real-world coordinates are obtained using an RTK marker. These coordinates are then compared with the coordinates in the model to analyze errors in the generated 3D model, including errors in planar coordinates and elevation. The distribution of the checking points is illustrated in Figure 8. Table 4 shows the comparison of the coordinates on the model and measured. It can be observed that the average absolute error values in the horizontal direction are relatively close, with average errors of 3.1 centimeters and 2.2 centimeters in the northward and southward directions, respectively. However, the average error in the vertical direction is larger, approximately twice to triple that of the horizontal direction. Compared with the positioning accuracy evaluated by Zhao for 3D photogrammetric reconstruction, it can be observed that the average horizontal-positioning error is around 3.5 centimeters and the average vertical-positioning error is around 6.5 centimeters, which is similar to the positioning accuracy in this paper [29]. Consequently, the errors

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Table 4: Error comparison between the RTK and model coordinates.

Nos.	RTK coordinates of CPs (N/E/Z) (m)	Model coordinates of CPs (N/E/Z) (m)	Error (N/S/Z) (cm)
CP1	654583.039 3536746.335 10.037	654583.005 3536746.353 9.941	3.4/−1.8/9.6
CP2	654561.892 3536730.986 10.179	654561.946 3536731.014 10.138	−5.4/−2.8/4.1
CP3	654119.623 3537182.779 12.668	654119.657 3537182.788 12.618	−3.4/−0.9/5.0
CP4	654097.247 3537160.323 12.653	654097.255 3537160.278 12.592	0.8/4.5/6.1
CP5	653461.794 3537790.680 9.533	653461.776 3537790.666 9.461	1.8/1.4/7.2
CP6	653453.416 3537781.560 9.631	653453.380 3537781.576 9.538	3.6/−1.6/9.3
Average of the absolute value			3.1/2.2/6.9

primarily concentrate on the vertical direction, meeting the requirement for centimeter-level model positioning accuracy.

The images collected by different flight paths were used to reconstruct 3D model of the pier, and the fine model is shown in Figure 12(b). A steel guardrail in the model is enlarged and its reconstruction precision is analyzed. The three-dimensional modeling diagrams of steel rails for route 1, route 2-3, and route 3 are, respectively, shown in Figures 12(c), 12(f), and 12(g), with a forward overlap rate of 80% and a side overlap rate of 70% for all three routes. It can be found that with the increase of the route distance, the reconstruction precision of the guardrail decreases. The steel guardrail of route 1 is better reconstructed, while the guardrail of route 3 is partially reconstructed. Figures 12(d), 12(e), and 12(f) demonstrate the modeling effect of the 20-meter route, whose route overlap rate is 60%, 70%, and 80% and side overlap rate is 50%, 60%, and 70%, respectively. It can be found that the reconstruction effect of the three models is quite different in the aspect of steel guardrail. The higher the overlap, the higher the integrity of guardrail reconstruction. According to the model reconstruction result, the GSD of Figures 12(c), 12(f), and 12(g) are 3.2 mm/pixel, 4.9 mm/pixel, and 7.2 mm/pixel, respectively.

5. Damage Detection and Localiztion Based on the 3D Model

5.1. Damage Detection Based on YOLOv7. In this study, a multidamage dataset of the Yangtze River Bridge was established, with 571 pictures captured by UAVs. The dataset consists of four category labels, including steel rust, concrete patch, corroded bolt, and clean bolt, with label amounts of 514, 390, 4209, and 14171, respectively. To address the imbalance between categories, label category weight and label image weight methods are employed to adjust the training process for YOLOv7.

During the regression process, the accuracy of predicting bounding box coordinates directly impacts the matching accuracy of damage features detected by the SIFT detector, thereby influencing the precision of damage coordinates in 3D mapping. In object detection, each target should be surrounded by a unique bounding box theoretically, while multiple prediction boxes encompass the same target. To tackle this challenge, a nonmaximum suppression (NMS) algorithm is employed to filter the redundant prediction boxes. The procedure for nonmaximum suppression is as follows [68]. First, prediction boxes are sorted by the confidence value, prioritizing those with the highest confidence. Subsequently, the prediction box with the highest confidence is selected as the starting point. Next, the intersection over union (IoU) ratio is computed between this chosen prediction box and the remaining ones in sequence, following the method outlined by Wu et al. [69]. If the IoU ratio surpasses the specified threshold, the remaining prediction boxes are filtered out. This cycle iterates until all remaining prediction boxes fall below the IoU threshold. However, for the same target, the NMS algorithm screens out similar prediction boxes with higher IoU, not suitable for intensive target detection with overlap.

For mutidamage model of YOLOv7, the average precision (AP) under different IoU threshold has been analysis, as shown in Figure 13. We set the IoU value range to [0.1, 0.9], with a step size of 0.1 and calculate the AP for each class in the dataset. The AP of steel rust exhibits a sharp decline when the IoU threshold exceeds 0.7. For concrete patch, corroded bolt, and clean bolt, their AP values begin to decrease as the IoU threshold surpasses 0.8. Specifically, concrete patch sees a slight decline at an IoU threshold of 0.9, whereas both corroded bolt and clean bolt experience

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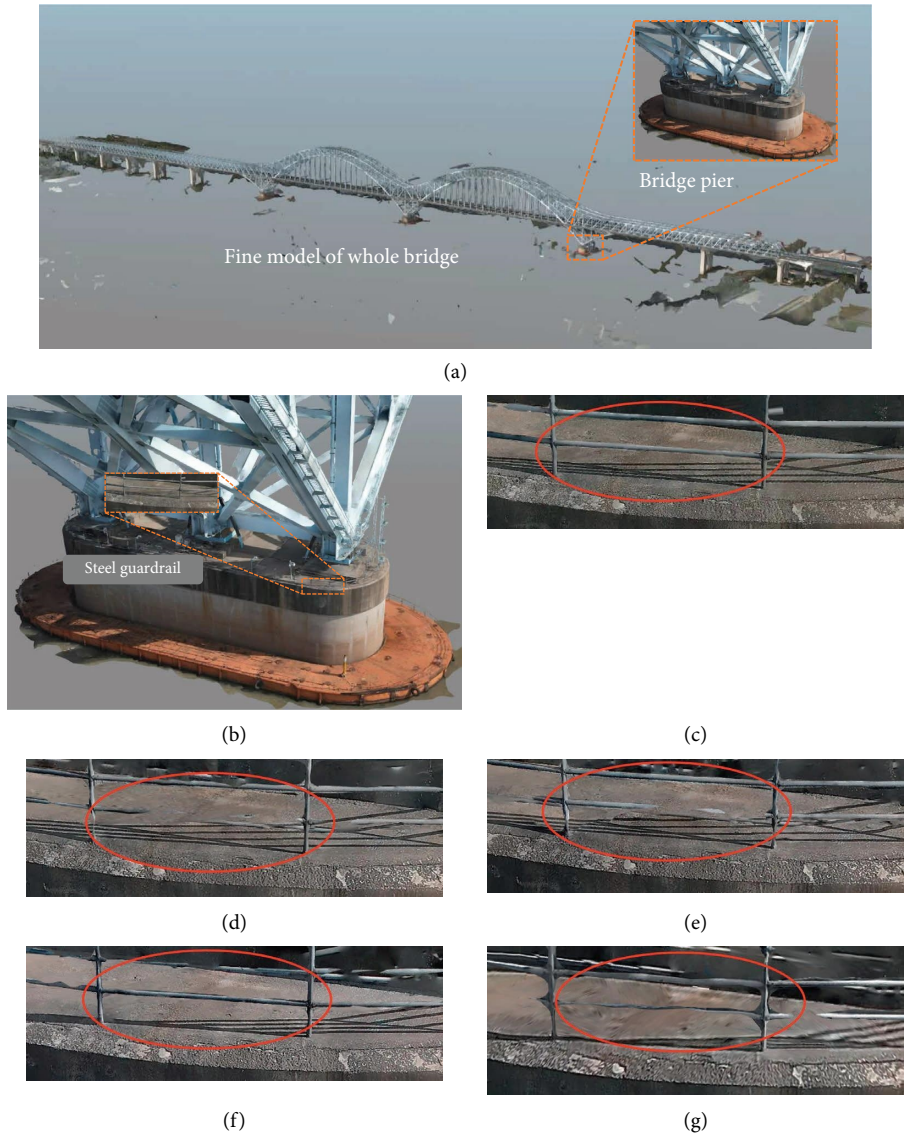


FIGURE 12: Comparison of pier models for different routes. (a) Fine model of the whole bridge. (b) Pier reconstruction model. (c) Route 1. (d) Route 2-1. (e) Route 2-2. (f) Route 2-3. (g) Route 3.

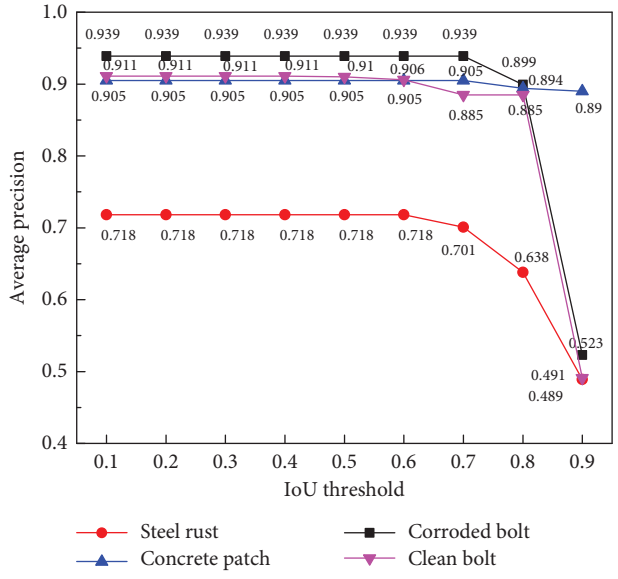
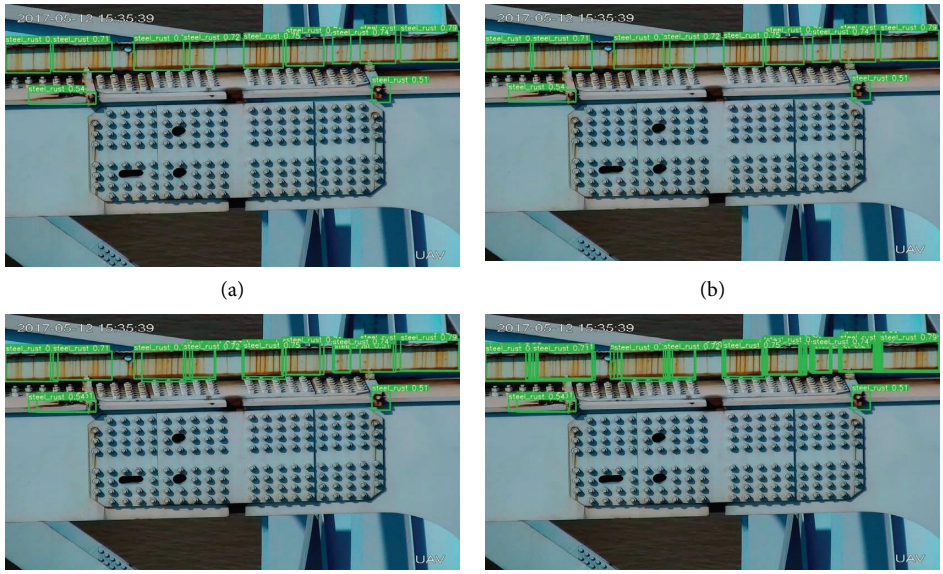


FIGURE 13: Average precision under different IoU threshold.

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(a) (b) (c) (d)

Figure 14: Detection results of steel rust under different IoU thresholds. (a) IoU = 0.3. (b) IoU = 0.5. (c) IoU = 0.7. (d) IoU = 0.9.

a significant decrease at this threshold. Consequently, in terms of AP, it is advisable to set the IoU threshold below 0.7. This recommendation stems from the observation that steel rust, characterized by its irregular shape, is notably more sensitive to overlap compared to concrete patch, corroded bolt, and clean bolt. Moreover, damages such as concrete patch, corroded bolt, and clean bolt exhibit a lower probability of overlap in comparison to steel rust.

Under different IoU thresholds, the detection results of damages tend to be variable. The steel rust was taken as example, as shown in Figure 14. When the IoU threshold is set to 0.3 or 0.5, only a few bounding boxes of steel rust overlap, and the overlap rate is small. If the IoU threshold is set to 0.7, there exist more than two prediction boxes around the same rust area. As the IoU threshold is set to 0.9, a large number of redundant detection boxes display. Considering the discrete characteristics of steel rust, it is appropriate to set the IoU threshold to below 0.3.

5.2. Coordinate Mapping between Damage Image and the 3D Reconstruction Model. According to Section 3.4, by feature matching and singular value decomposition of the matrix, the 3D spatial coordinates of the steel rust can be obtained. The 3D spatial coordinates follow the universal transverse mercator (UTM) grid system, which is a rectangular coordinate system. The origin of UTM is located at the intersection of the meridian and the equator in the center of the longitude region. Within this system, east-facing denotes the projected distance from the central meridian of the longitude region, serving as the positive direction of the X-axis in this study. Similarly, north-facing signifies the projected distance from the equator and represents the positive direction of the Y-axis. The positive Z-axis is perpendicular to the projection plane and opposite to the geocentric direction.

The positioning coordinates of the 3D reconstruction model were taken as the ground truth values, and the surface damage coordinates of by 3D mapping were adopted for absolute error analysis, as shown in Table 5. As the UAV flight distance increased, the absolute error of the 3D mapping coordinate enlarges, where the error in Z-axis is larger than that of other axes. As the flight distance is within 10 m, the error in the X-axis, Y-axis, and Z-axis is controlled around 0.2 m, 0.2 m, and 0.5 m, respectively. When the flight distance is 20 m, the horizontal and vertical localization accuracy of the 3D mapping coordinate is 0.5 m and 1 m, respectively. As the flight distance exceeds 20 m, the localization error accelerates rapidly.

In Figure 15, the relative error has been analyzed for 3D mapping accuracy of the damage. As the flight distance is 20 m, the relative error of X-axis and Y-axis is 2.5% and 2.3%, respectively, and the relative error of Z-axis is 6.1%. Therefore, it is recommended that the flight-shooting distance of the UAV be controlled within 20 m to achieve the meter-level accuracy.

5.3. Visualization of Damage Spatial Mapping. The 3D reconstruction model of the bridge is obtained in Section 3. According to the spatial mapping coordinates of the damage, the detailed information can be associated with the 3D model and visually displayed. The 3D GIS software was used to connect the damage type and detailed detection results at the corresponding damage position. In addition, the spatial information of the damage, including longitude, latitude, and elevation, was detailed in the software. Through visualization, the damage distribution of the bridge can be clearly displayed intuitively, and the damage information is detailed, such as steel rust and concrete patch.

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Table 5: Absolute error analysis of 3D mapping accuracy.

Flight distance (m)	Damage type	*X*-axis error (m)	*Y*-axis error (m)	*Z*-axis error (m)
10	Steel rust	0.22	0.21	0.53
20	Steel rust	0.49	0.46	1.22
30	Steel rust	1.35	1.86	3.56
40	Steel rust	3.96	4.55	8.92
50	Steel rust	6.59	7.32	15.6

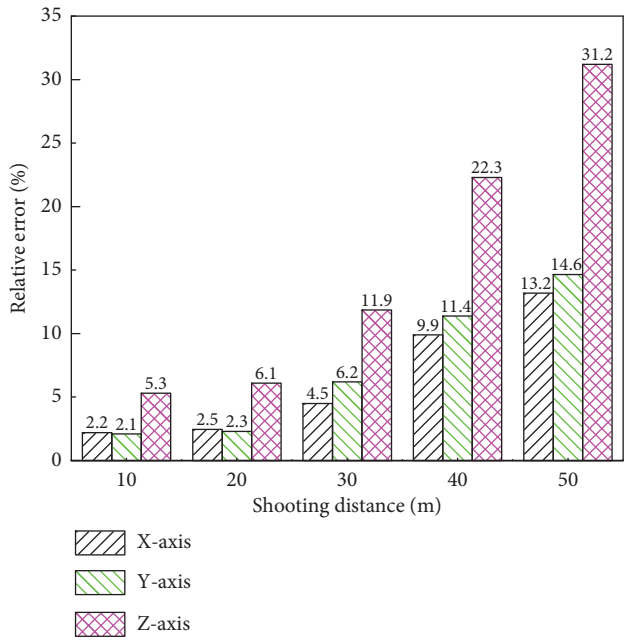


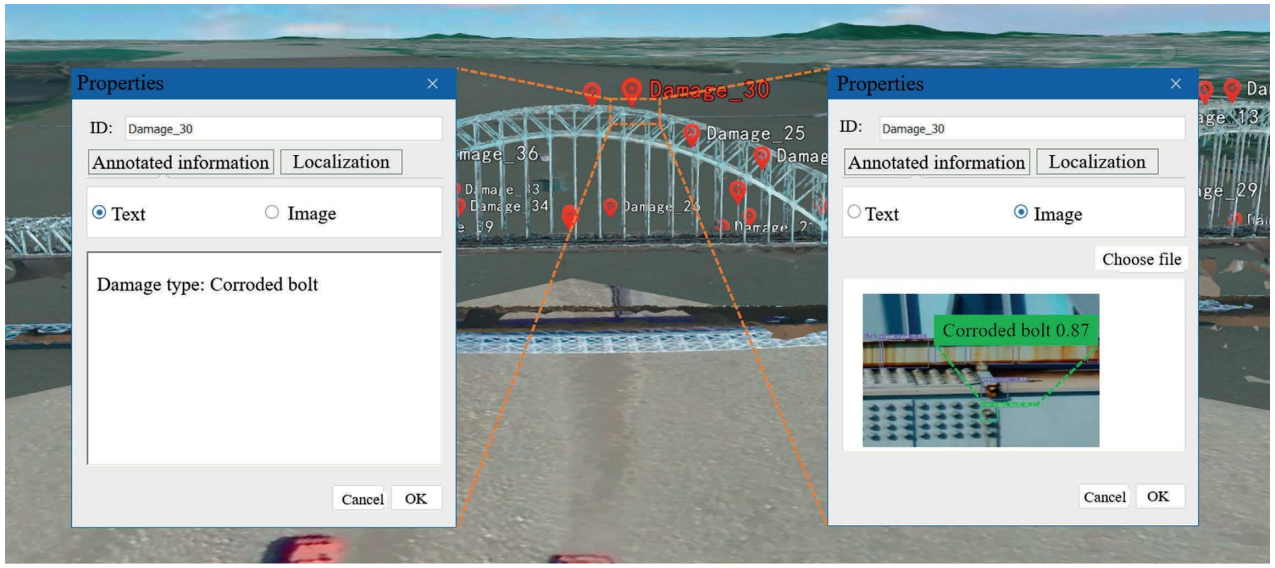
Figure 15: Relative error analysis of 3D mapping accuracy.



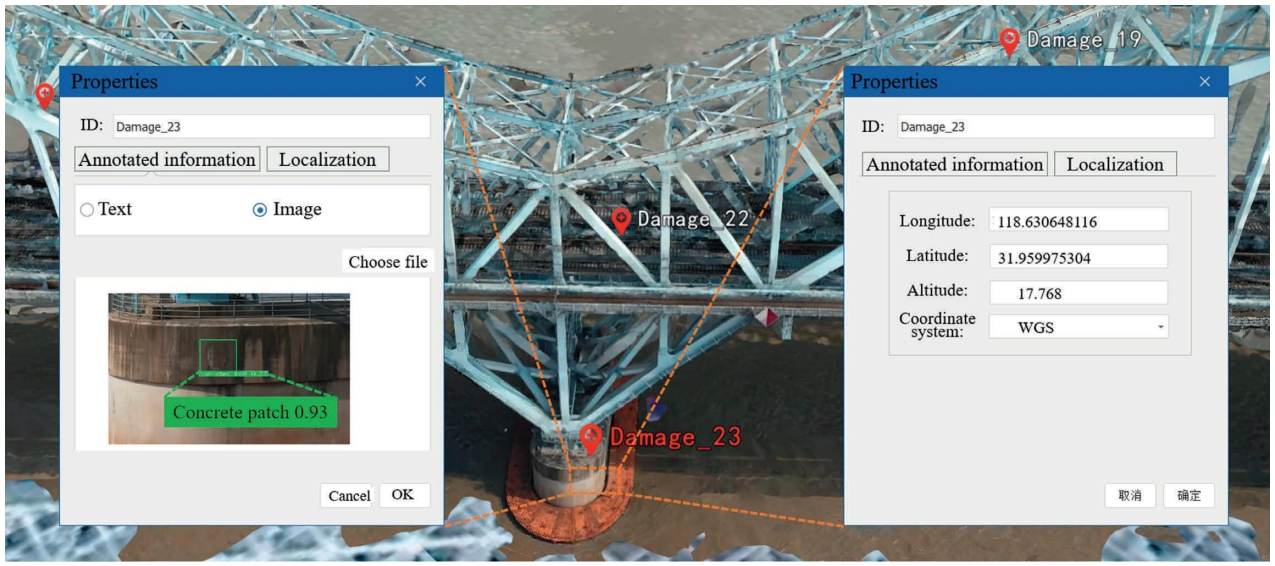
(a)

Figure 16: Continued.

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(b)



(c)

Figure 16: Visualization of damage 3D mapping. (a) Damage distribution along the model. (b) Corroded rust. (c) Concrete patch.

The distribution of damage along the model is depicted in Figure 16(a). It is evident that the damage is predominantly concentrated in the steel structure, particularly near the arch and the upper chord of the steel truss. Presently, there exist 52 surface damages marked statistically, that is, steel rust, corroded bolt, and concrete patch. The mapped model damage information of corroded bolt and concrete patch is illustrated in Figures 16(b) and 16(c), respectively. Upon clicking the button of the damage mark, the properties window appears. Within the annotated content text item, the damage type is displayed. Switching to the image item reveals the damage detection results. Moreover, in the localization information section of the properties window, the World Geodetic System (WGS) coordinates of the selected damage are presented. For instance, in Figure 16(b), Damage_30 is identified as a corroded bolt with a confidence value of 0.87. In Figure 16(c), Damage_23 is classified as a concrete patch, with the corresponding WGS coordinates provided (east longitude: 118.630648116°; north latitude: 31.959975304°; and elevation: 17.768 m).

The 3D reconstruction model, coupled with damage spatial mapping, facilitates the quantification and visualization of damage detection. This not only enables precise management plans for the bridge owner based on statistical damage data but also allows for the synchronous updating of damage information postdiagnosis and treatment into the model. Consequently, this integration supports ongoing bridge maintenance throughout its lifecycle.

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6. Conclusion

In this study, a three-phase method of surface damage detection and spatial localization is developed based on YOLO network and three-dimensional mapping. First, according to the preliminary bridge model, the detailed path of the UAV is planned, and the fine reconstruction of the bridge is realized by the SfM algorithm. Second, the YOLOv7 algorithm is employed to detect multiple damages, achieving image-level matching through the SIFT detector by utilizing the coordinates of the damage bounding boxes. Finally, based on the polar geometry constraint solving, the mapping from the damage pixel level to the 3D model is realized. The application and evaluation of this method were conducted on a Yangtze River bridge, and the main conclusions are as follows:

(1) The route planning path adopted consists of spiral circling of bridge piers and parallel flight of both sides of the whole bridge. The shooting distance of the UAV is controlled at 20 m, the forward overlap rate is greater is 80%, the side overlap rate is 70, and the GSD is 4.9 mm/pixel.

(2) As the IoU threshold exceeds 0.8, the detection accuracy of concrete patches, corroded bolt, and clean bolt begin to decline. Therefore, it is advisable to set the IoU threshold below 0.7. In addition, due to their discrete characteristics, the IoU threshold for steel rust should be set even lower, preferably below 0.3.

(3) The average absolute error values are 3.1 centimeters and 2.2 centimeters in the northward and eastward directions, respectively. However, the average error in the vertical direction is approximately two to three times greater than that in the horizontal direction. The reconstruction model of bridges achieves centimeter-level positioning accuracy.

(4) The horizontal and vertical localization accuracy of bridge damage is 0.5 m and 1 m, respectively. The relative error of positioning in X, Y, and Z direction is 2.5%, 2.3%, and 6.1%, respectively. The accuracy meets the requirements of 3D reconstruction model for damage positioning, thus providing valuable reference for bridge maintenance [6, 25].

Data Availability

The data used to support the findings of this study are available on request from the corresponding author.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

Acknowledgments

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